Air University Islamabad

FACULTY OF COMPUTING & ARTIFICIAL INTELLIGENCE

Department of Creative Technologies

Final Project

Introduction to Data Science

BSDS 3-A

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Contents:

Data Driven Analysis of Pakistan’s Education Sector

Introduction

Pakistan’s education sector faces persistent challenges, including low literacy rates, gender disparities in enrollment, and inefficiencies in budget allocation. This project aims to provide a comprehensive analysis of Pakistan’s education sector by uncovering critical trends and patterns.

Data Collection:

The data used in this report is sourced from the Pakistan Social and Living Standards Measurement (PLSM) Survey and the Labor Force Survey conducted by PBS (Pakistan Bureau of Statistics). These sources are compiled and published in the Education Report by the Finance Division in the form of a PDF document, which we have utilized for analysis.
https://www.finance.gov.pk/survey/chapters_23/10_Education.pdf

Metadata:

Expenditure Table

Column Name	Description
Year	Year of data collection
Region	Region or province where data was collected. (Federal, Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan, Pakistan.)
Current Expenditure	Expenditure on current/operational costs in education.
Total Expenditure	Total expenditure, combining current and development costs.
Percent of GDP (2015-16 Base)	Percentage of total economic output (GDP) spent on education, using 2015-16 as the base year for inflation-adjusted comparisons.

Facility Table

Column Name	Description
Region	Region or provinces where facilities are located.
Facility	Type of facilities (Access to Electricity, Availability of Drinking water, Availability of Toilet, Availability of Boundary Wall)
Primary	Percentage of primary-level institutions having these facilities.
Middle	Percentage of middle-level institutions having these facilities.
High	Percentage of high-level institutions having these facilities.
Higher Sec.	Percentage of Higher Sec.-level institutions having these facilities.
Percentage Total	Percentage distribution of facilities among levels.

Literacy Rates Table

Column Name	Description
Province/Area	Province or area where literacy data is reported.
Year	Year of data collection.
Male	Literacy rate among males (percentage).
Female	Literacy rate among females (percentage).
Total	Overall literacy rate (percentage).

Institutes, Enrollments and Teachers Table

Column Name	Description
Year	Year of data collection.
Institution Type	Type of educational institution

Gender	Gender distribution in institutions (Male/Female).
InstitutesNumber	Number of educational Institutes.
Enrollments	Number of student enrollments.
Teachers	Number of teachers available.

Concepts

Descriptive Statistics

- Summarizes enrollments, literacy rates, and facility availability across genders, regions, and education stages.
- Computes the Gender Parity Index (GPI) as the ratio of female to male values for enrollments and literacy rates.
- Measures percentages of schools meeting basic facility criteria (e.g., electricity, water, toilets, boundary walls).
- Summarizes spending levels for analysis.
- **Filtering and Summing** - Focuses on the latest year and specific regions for expenditure analysis.

Inferential Statistics (Probability and Ratios)

- Evaluates gender proportions in enrollments and literacy rates.
- Tracks the percentage of students transitioning between education stages (e.g., Primary to Middle Schools) by gender by aggregating data to calculate mean transition ratios over multiple years (2010–2023) for a comprehensive view.
- Measures the number of students per teacher, indicating resource availability.

Categorical Data Analysis

- Filters data by urban/rural areas and provinces for regional insights.
- Uses grouped bar charts to compare facility access across regions and education levels.
- Employs melted data format for easier comparison across education stages.
- Groups data by **regions** to examine expenditure patterns.

Hypothesis Testing

- Applies Chi-Square Test of Independence to determine if differences in male and female enrollments are statistically significant.
- Generates a p-value (significant if < 0.05) to validate differences.

Time Series Analysis

- Analyzes trends in gender-based enrollments over time across different education stages (Primary, Secondary, Higher Secondary, etc.).
- Groups data by year, institution type, and gender to identify patterns in enrollment trends.
- Analyzes changes in education spending over time to identify patterns.
- Tracks changes in spending over time.

Linear Regression Analysis

- Examines how spending affects teacher numbers, using a straight-line formula to show changes and measure relationship strength (R-squared).
- Examines the relationship between spending and facility access, predicting changes using a straight-line formula.

Key Questions

1. How well does the Gender Parity Index (GPI) reflect the balance between male and female participation in education and literacy?
2. Are schools providing basic facilities, and how close are they to meeting the 90% target?
3. Is there a significant difference in male and female school enrollments, and how have these enrollments changed over time across school types?
4. What percentage of males and females transition between education levels, and how does this differ?
5. How has the percentage of GDP and total expenditure on education varied over the years?
6. What are the regional contributions to national education expenditure, both in 2022-23 and over the years?
7. How do student-teacher ratios vary across institution types and years, and how do they compare to the target ratio of 20:1?
8. How has the student-teacher ratio changed over time across different institution types?
9. How does development expenditure affect the number of teachers in different regions, and which regions use it most efficiently?
10. How does development spending impact access to educational facilities, and which regions are most efficient in using it?

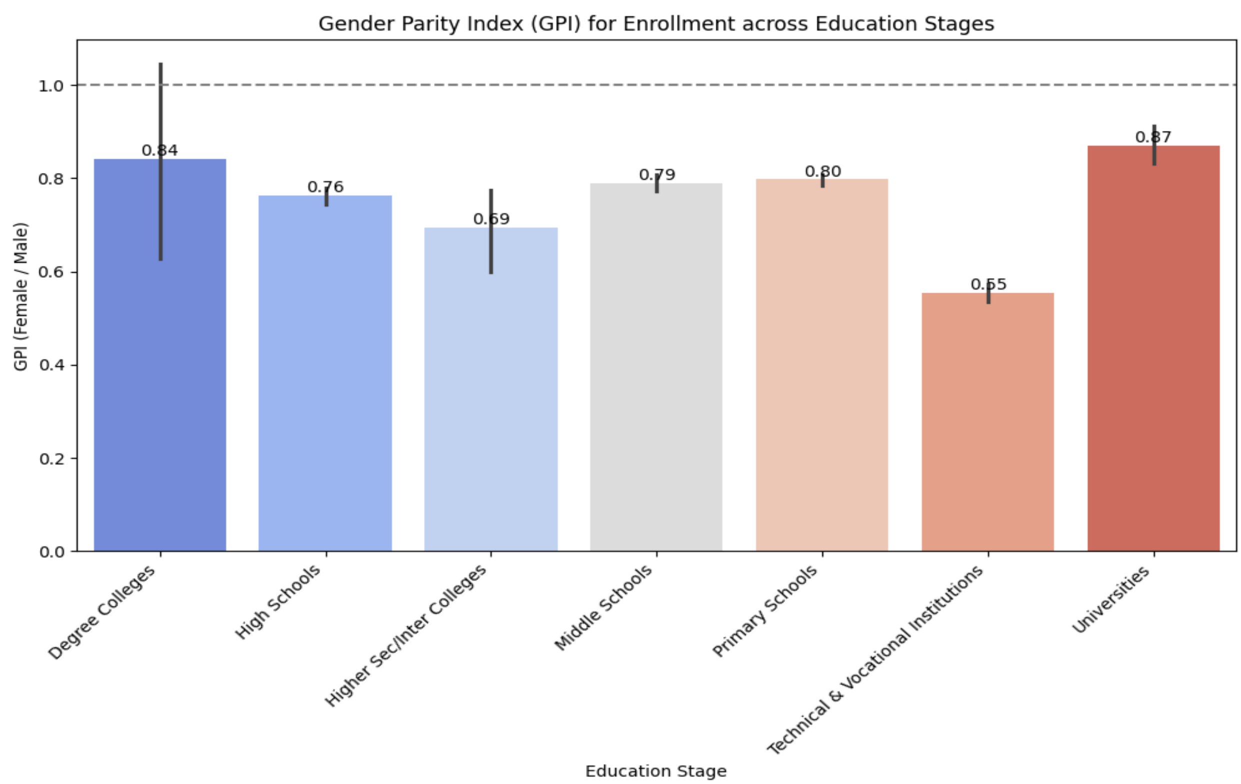
Visual Analysis of State of Education in Pakistan

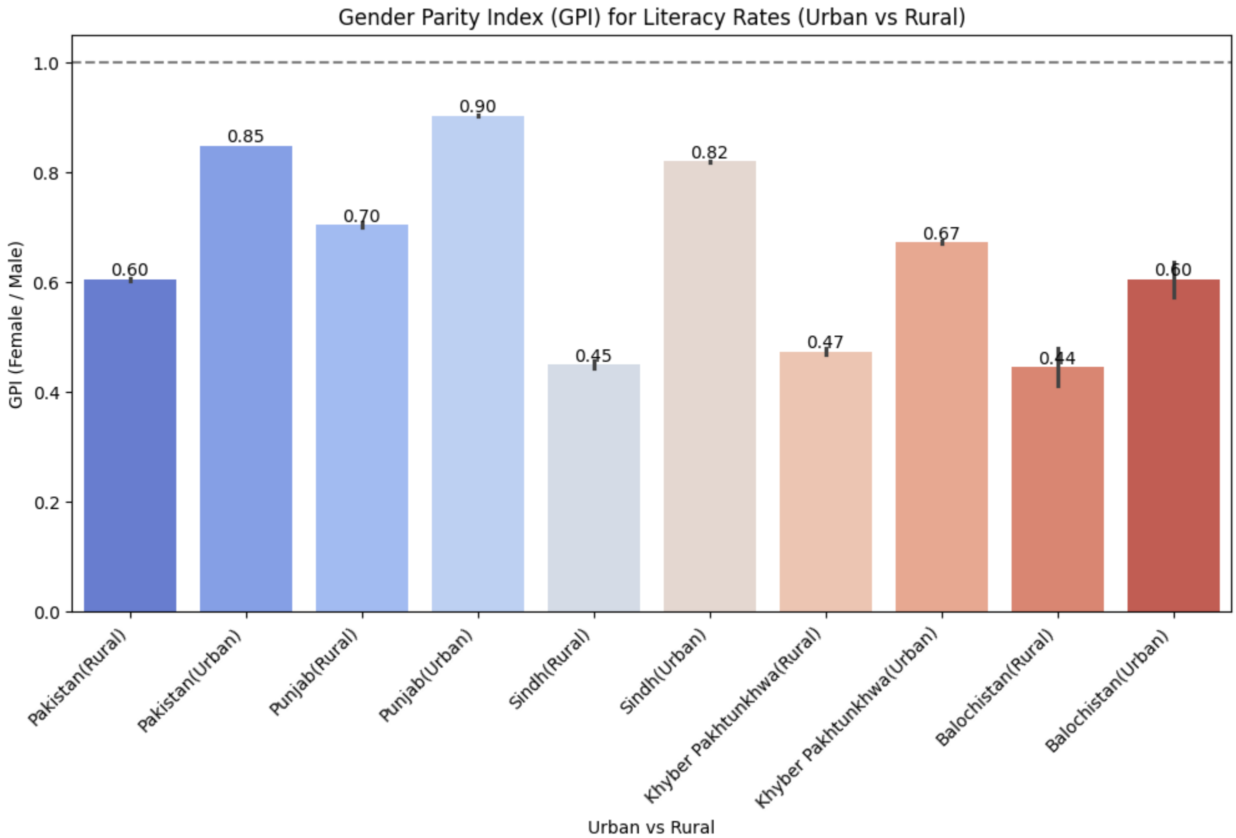
1. SDG 4 Indicator 4.5.1: Parity indices (female/male, rural/urban, etc.) for all education indicators that can be disaggregated

- GPI measures gender equality in education by comparing the number of females to males in enrollment or literacy rates

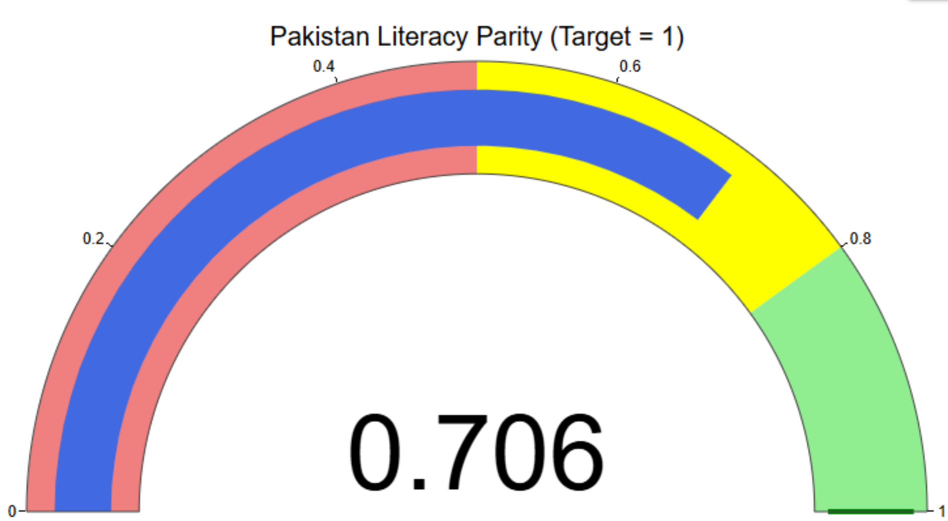
Insight:

- Gender parity in education enrollment in Pakistan is high in primary and middle schools but declines significantly in higher education, especially in universities and technical institutions.
- Gender parity in literacy rates is generally higher in urban areas than rural areas across all provinces in Pakistan, with Sindh and Balochistan showing the largest urban-rural disparity.





Tracking gender equality in literacy rates in Pakistan

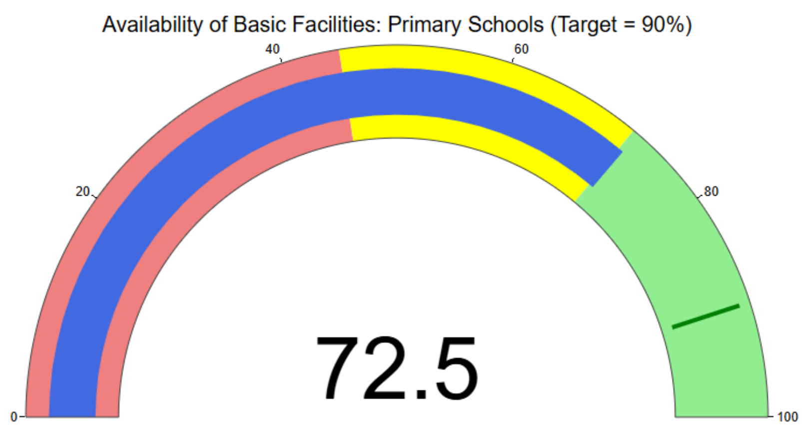


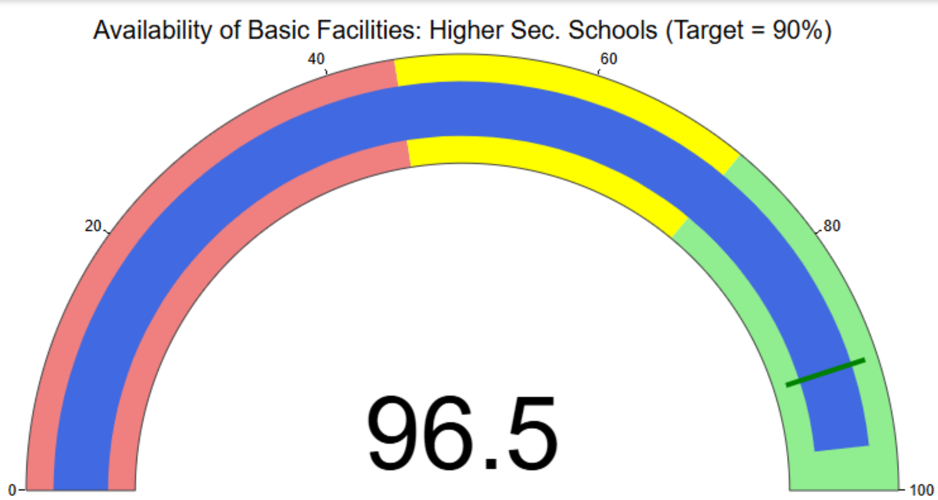
2. SDG4 Indicator 4.a.1: Proportion of schools with access to basic services according to 2021-22 year data

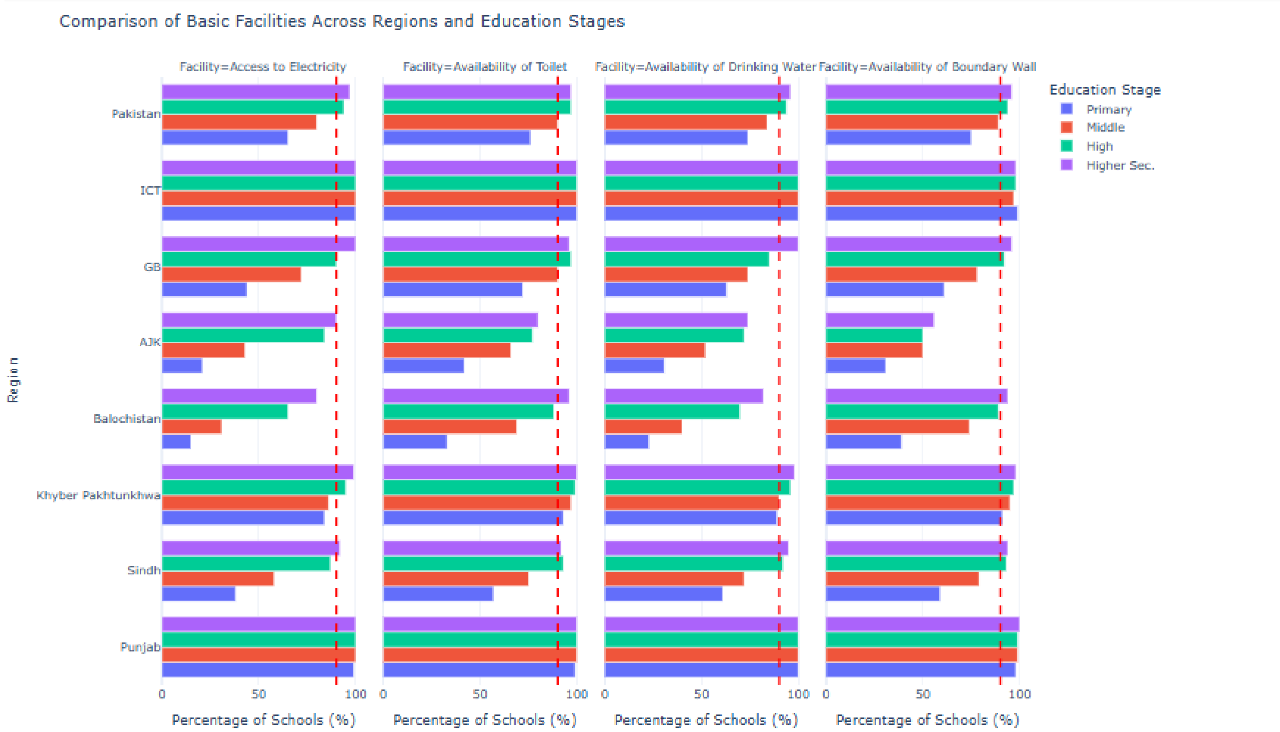
Target line at 90% to visually assess which regions and education stages meet or fall short of the desired facility availability.

Insight:

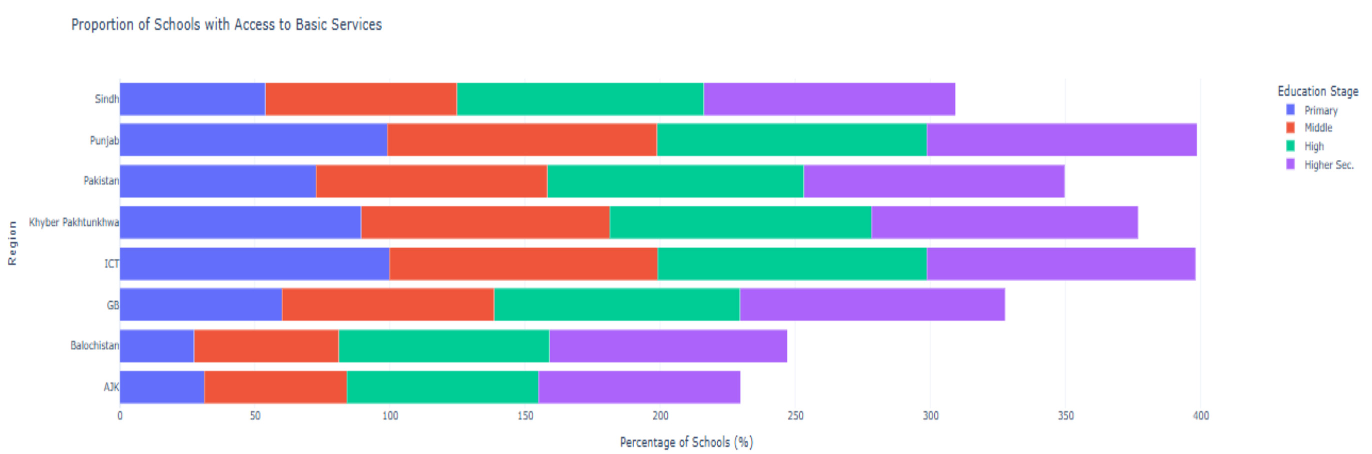
- Schools in Punjab and Sindh have the best access to basic facilities.
- Access to electricity and drinking water is generally high across regions.
- Toilet availability and boundary walls are areas of concern, especially in schools outside Punjab and Sindh.
- Higher Secondary schools tend to have better facilities compared to Primary schools.







Tracking Availability of Facilities across region.

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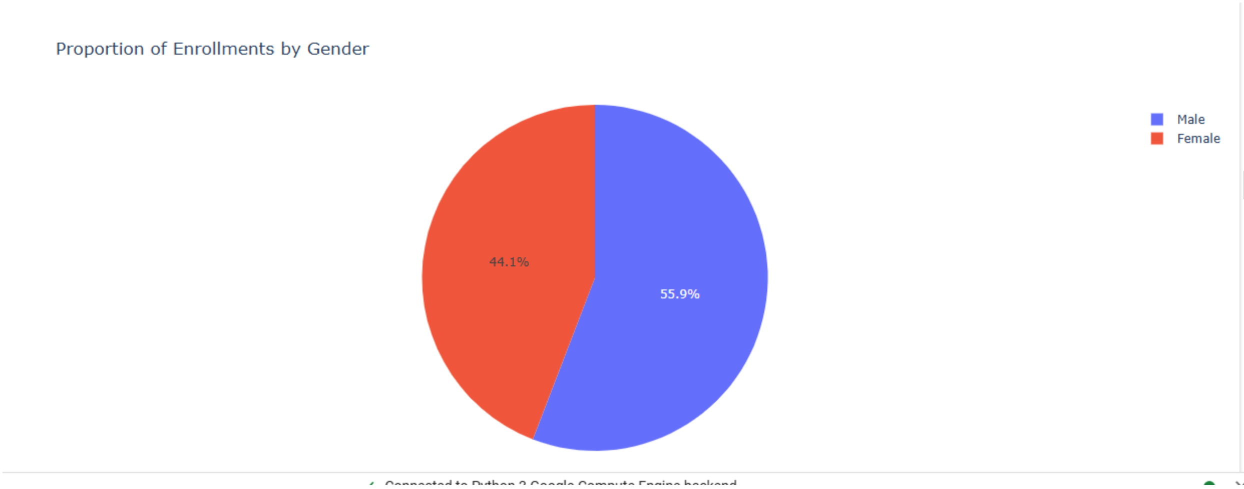
import pandas as pd
```

3. Gender Enrollment Analysis and Chi-Square Test

Concept:

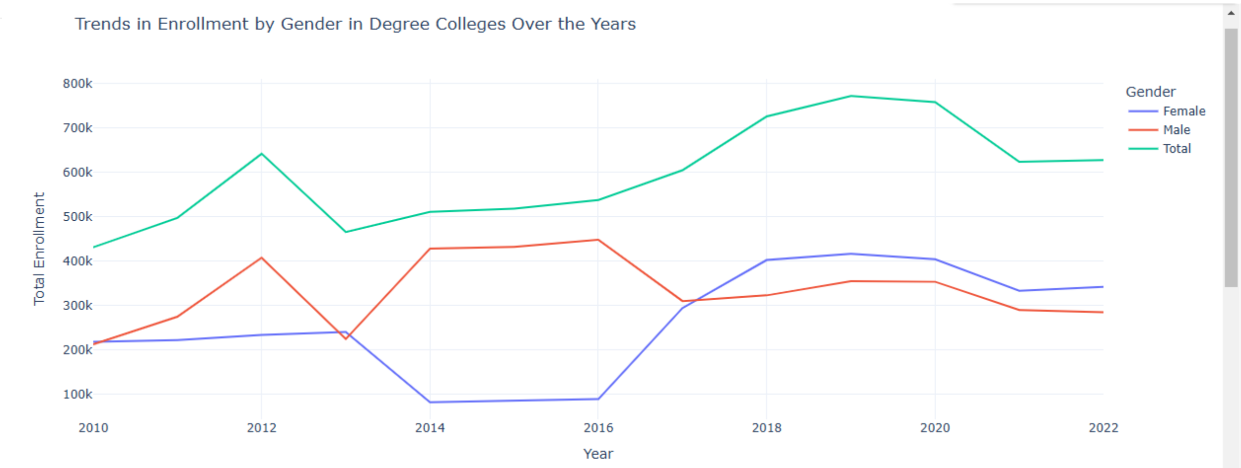
- Hypothesis Testing:
- **Null Hypothesis (H_0):** The proportion of female enrollments is the same as that of male enrollments.
- **Alternative Hypothesis (H_1):** The proportion of female enrollments significantly differs from male enrollments.
- Chi-Square Test of Independence:
  - Used to test if there's a **significant difference** in the enrollments by gender.
  - Based on the **contingency table** that compares the observed vs. expected frequencies.
- Chi-Square Statistic:
- The test produces a **p-value** that helps determine statistical significance. A **p-value < 0.05** suggests that the difference is statistically significant.
- Proportions and Ratios:
- **Enrollment Proportions:** Summarizes the total number of enrollments for each gender.

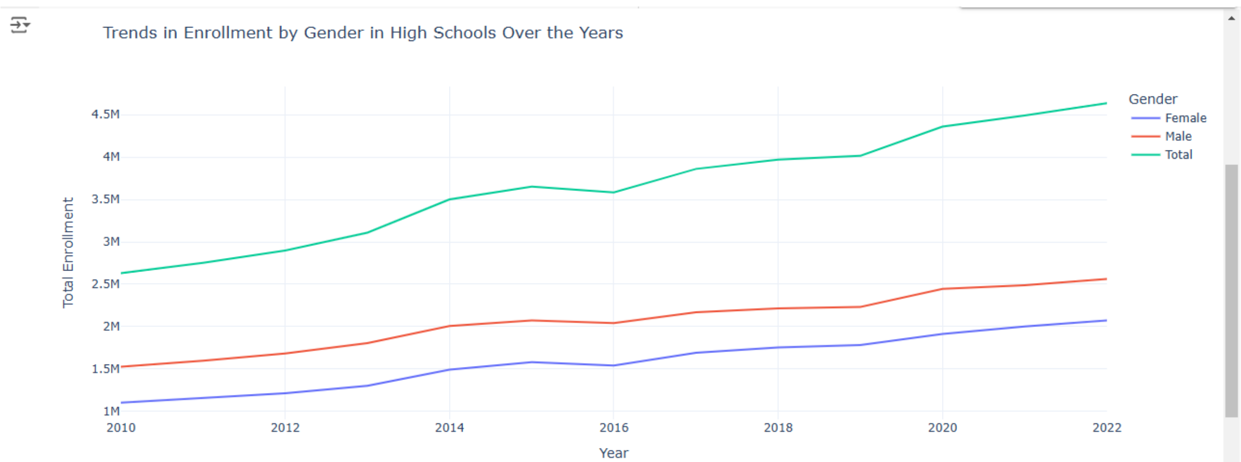
Insight: The pie chart shows that female students make up a slightly higher proportion of enrollments (55.9%) compared to male students (44.1%).

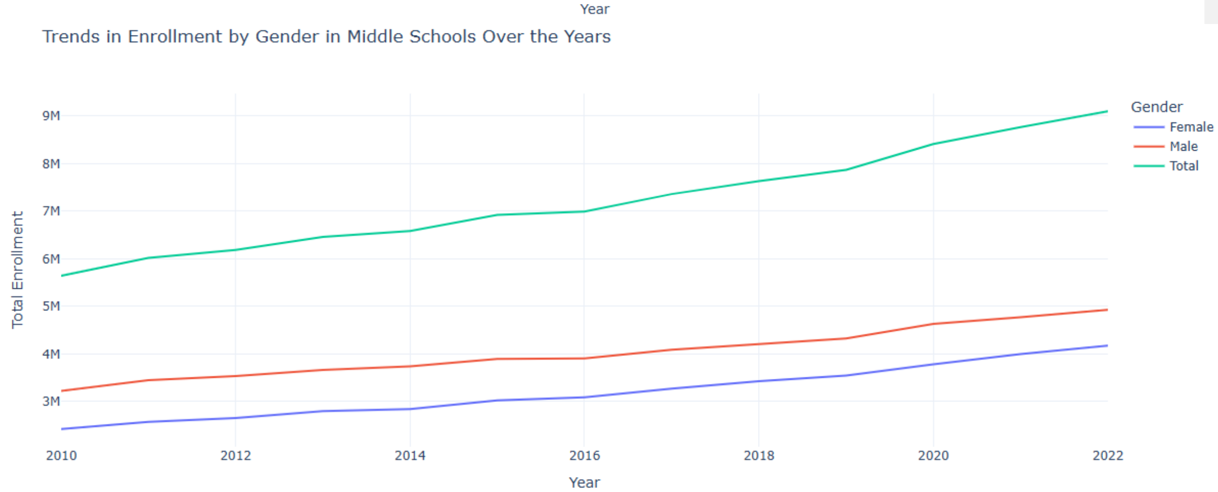


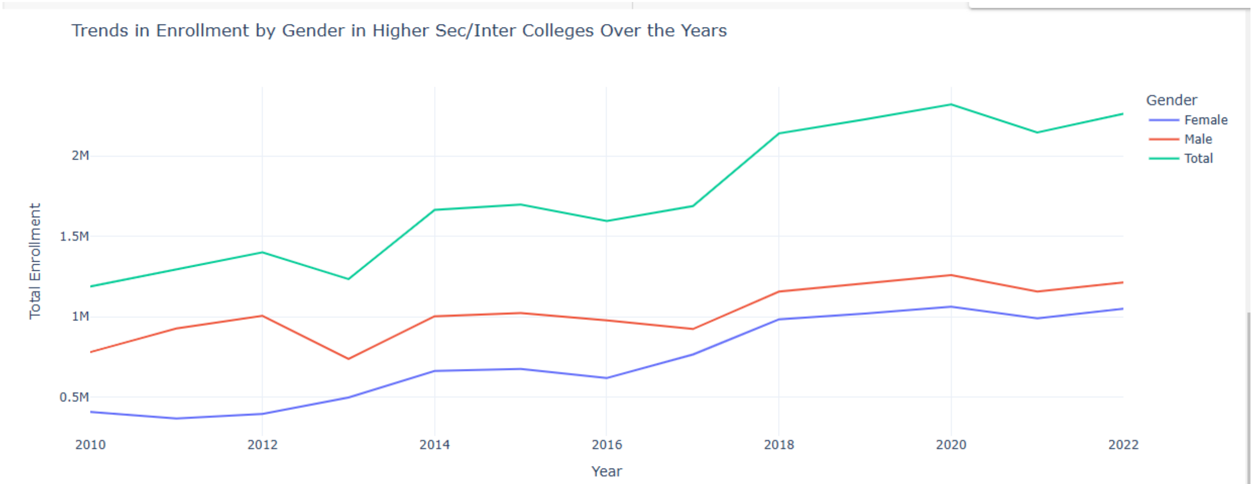
4. Gender-Based Enrollment Trends Over the Years: Analysis by Institution Type and Region

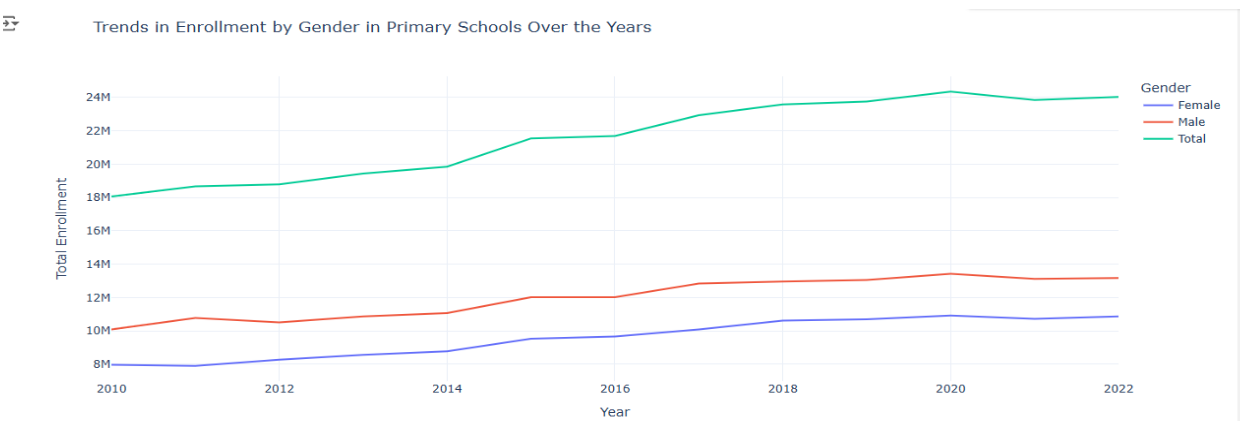
Insight: Female trends are lower across each institution type

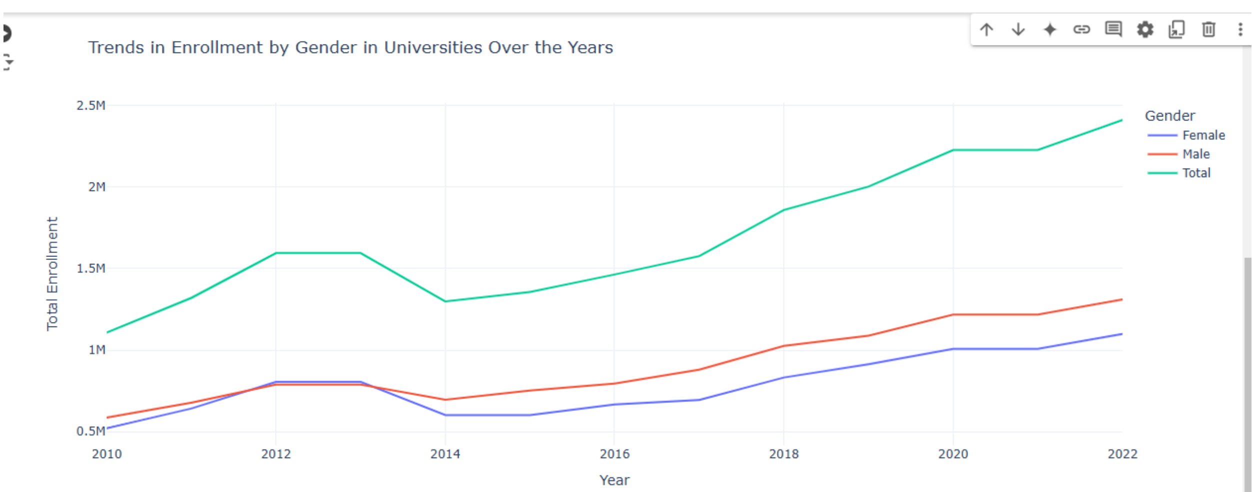







Trends in Enrollment by Gender in Higher Sec/Inter Colleges Over the Years
Gender
Female
Male
Total
Total Enrollment
2M
1.5M
1M
0.5M
2010
2012
2014
2016
2018
2020
2022
Year

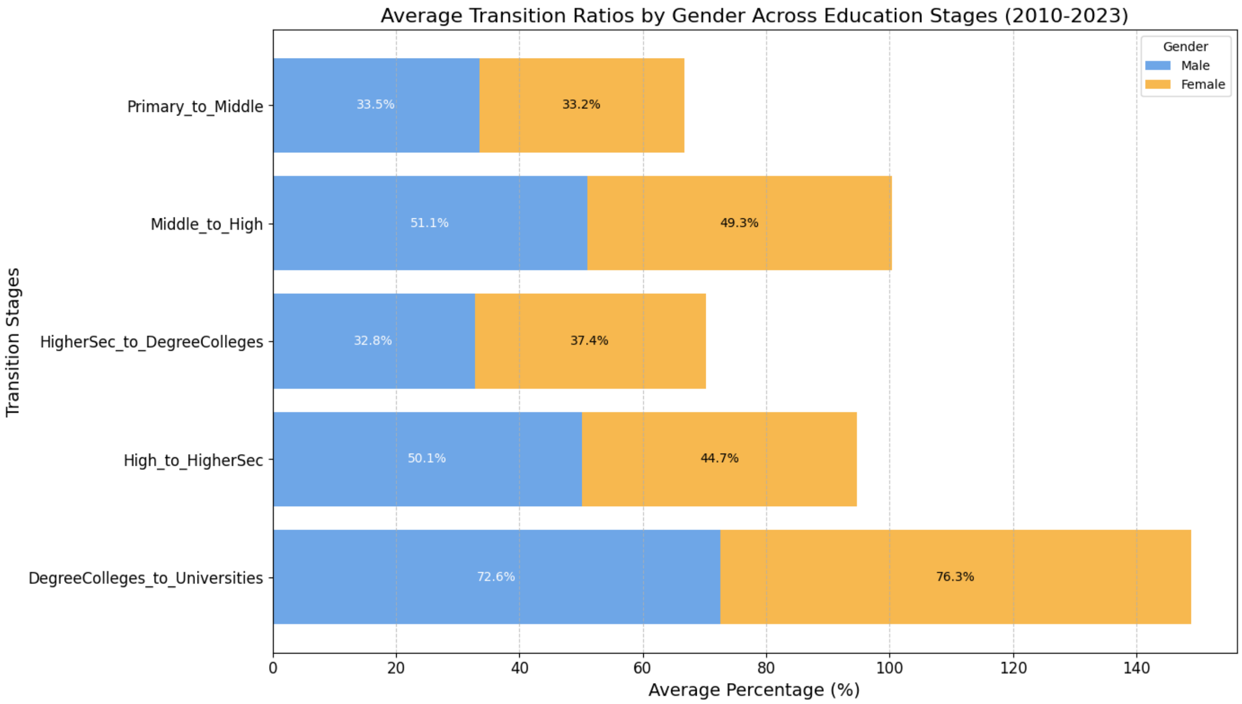

Trends in Enrollment by Gender in Primary Schools Over the Years
Gender
Female
Male
Total
Total Enrollment
24M
22M
20M
18M
16M
14M
12M
10M
8M
2010
2012
2014
2016
2018
2020
2022
Year


Trends in Enrollment by Gender in Universities Over the Years
Gender
Female
Male
Total
Total Enrollment
2.5M
2M
1.5M
1M
0.5M
2010
2012
2014
2016
2018
2020
2022
Year

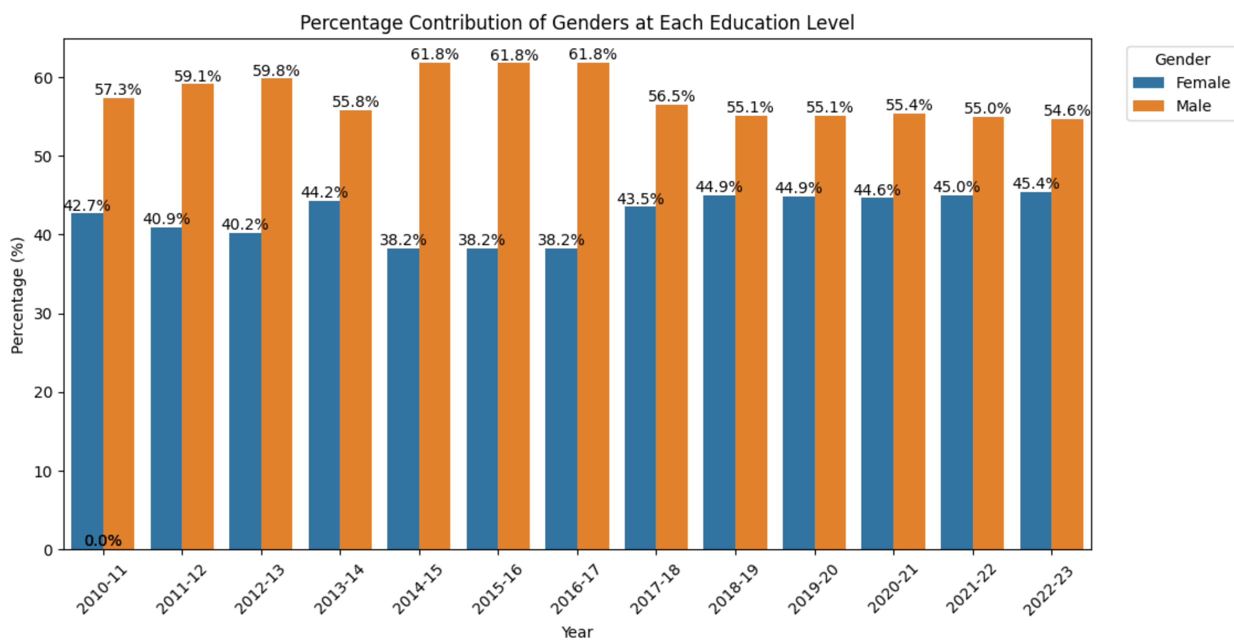
5. Transition Ratios:

The **mean percentage** across all years (2010-2023) is calculated to give an overall view of the gender-based transition ratios. This reduces the noise from year-to-year variations.

Insight Females are more likely to continue their education than males, especially in higher education, but males are more likely to transition from primary to secondary school.



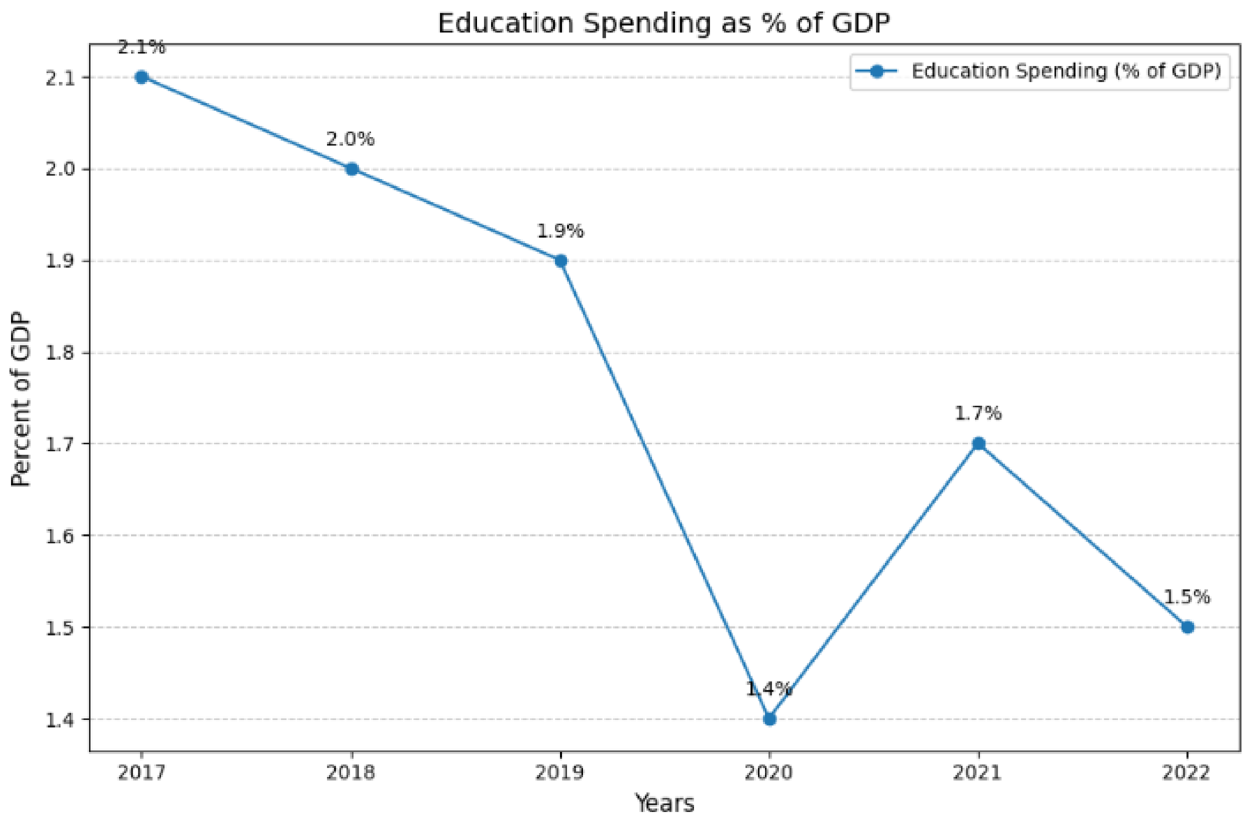
6. Yearly Contribution of Gender at Education Levels:



Education Expenditures

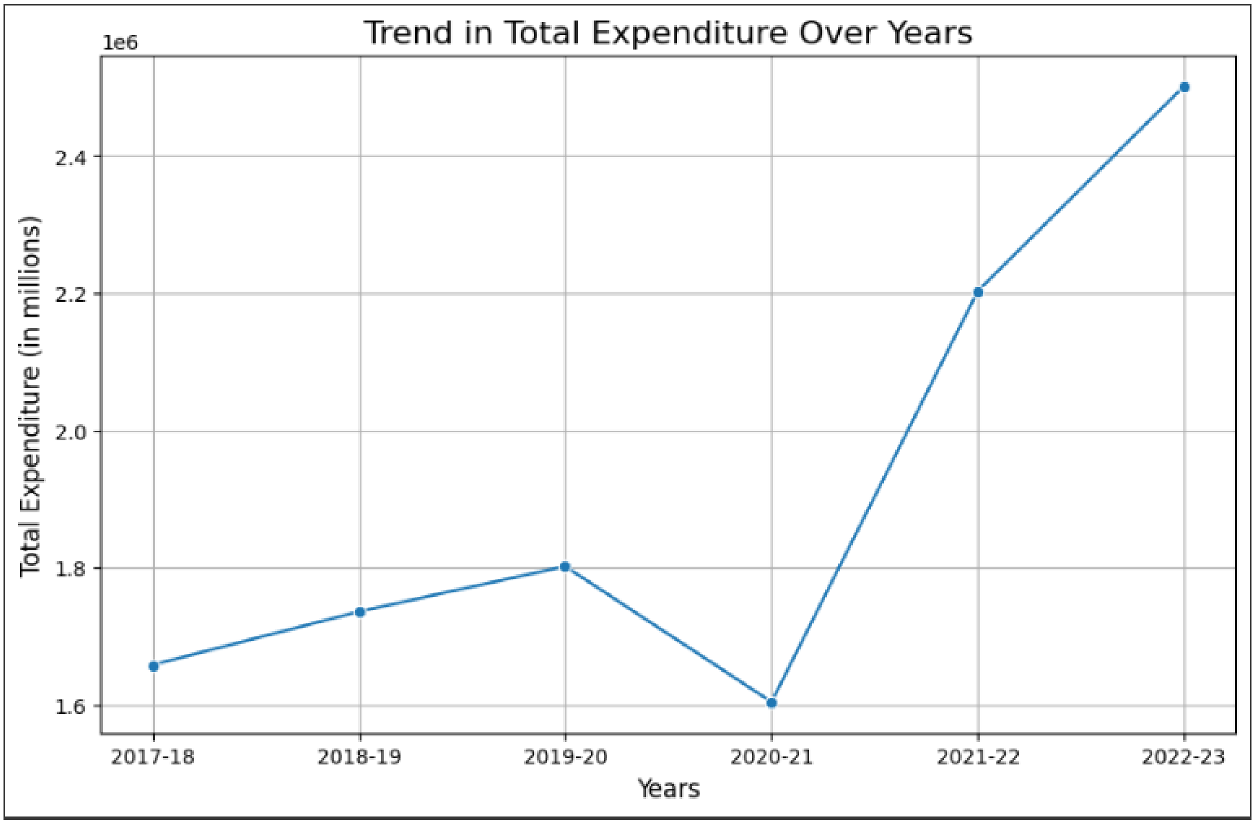
Trend of Education Spending as %age of GDP

Insight: The line graph shows a decline in education spending as a percentage of GDP from 2017 to 2020, followed by a slight increase in 2021 and a further decline in 2022.



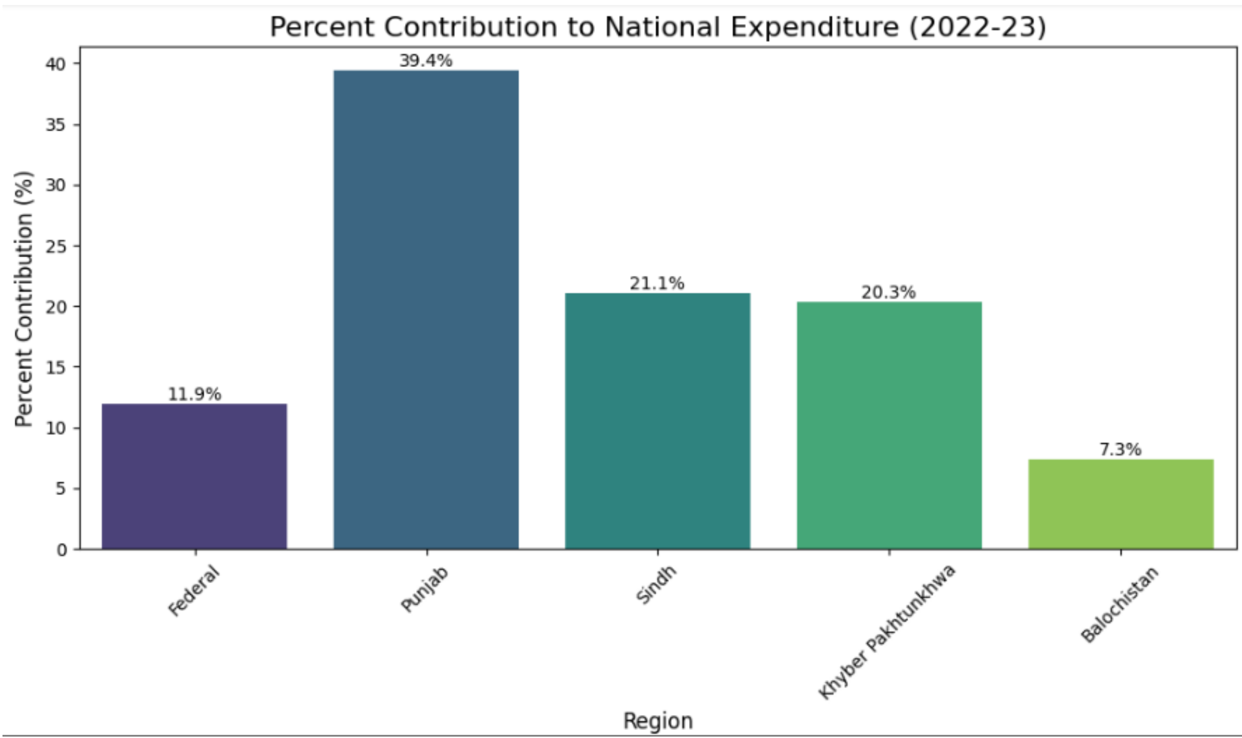
Trend of Total Expenditures over the Years

Insight: The line graph shows a fluctuating but overall increasing trend in educational expenditure across Pakistan from 2017-18 to 2022-23. A sharp dip in 2020-21 is observed, possibly due to the pandemic.



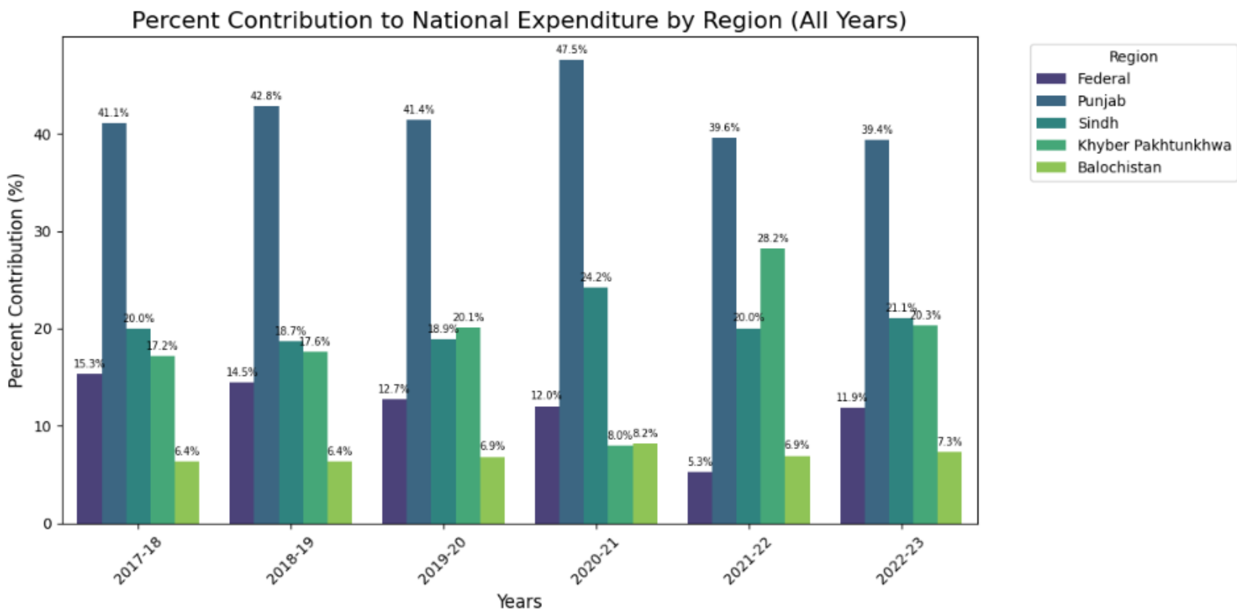
Percent Contribution to National Expenditure by Regions (Latest Year)

Insight: Punjab leads national expenditure with 39.4%, while Balochistan contributes the least at 7.3%, highlighting regional spending disparities.



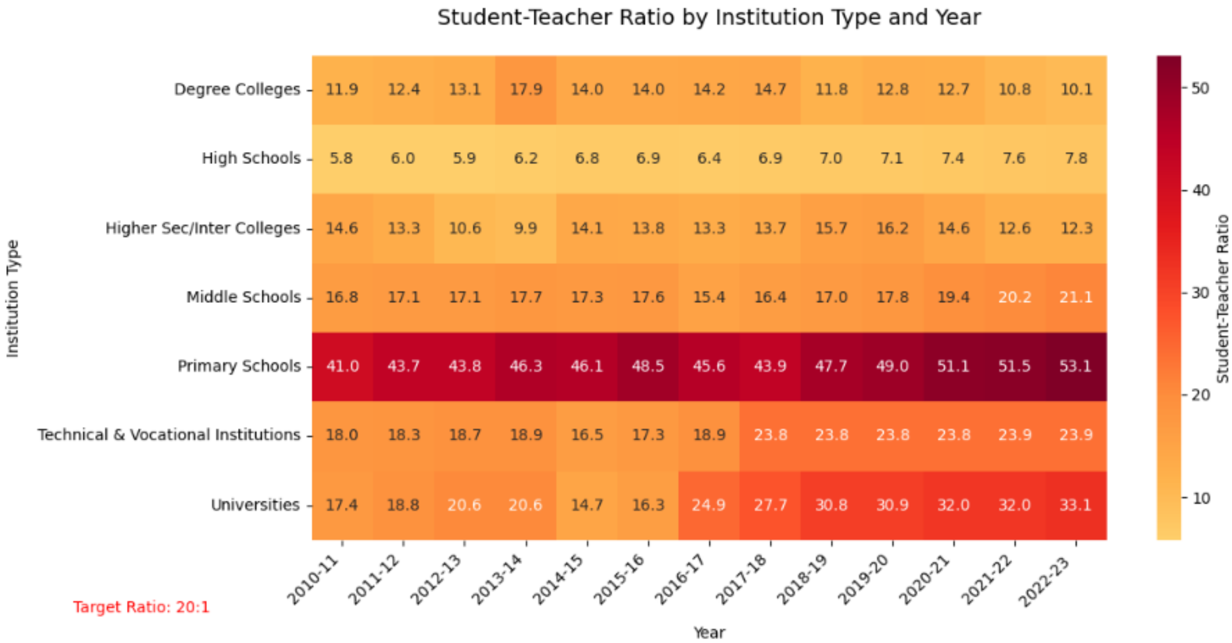
Percent Contribution to National Expenditure by Regions (All Year)

Insight: Punjab dominates national expenditure yearly, peaking at 47.5% in 2020-21, while Sindh, Khyber Pakhtunkhwa, Federal, and Balochistan show smaller and consistent contributions, highlighting regional disparities.



Student-Teacher Ratio

Insight: The heatmap shows that primary schools have the highest student-teacher ratios (41:1 to 53.1:1), far exceeding the target of 20:1, while universities and colleges maintain lower, more balanced ratios. Middle schools also show a rising trend, highlighting critical gaps in teacher availability at foundational education levels.

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Average Ratios:
Institution Type
Degree Colleges                        13.1
High Schools                            6.8
Higher Sec/Inter Colleges              13.4
Middle Schools                         17.8
Primary Schools                        47.0
Technical & Vocational Institutions    20.7
Universities                           24.6
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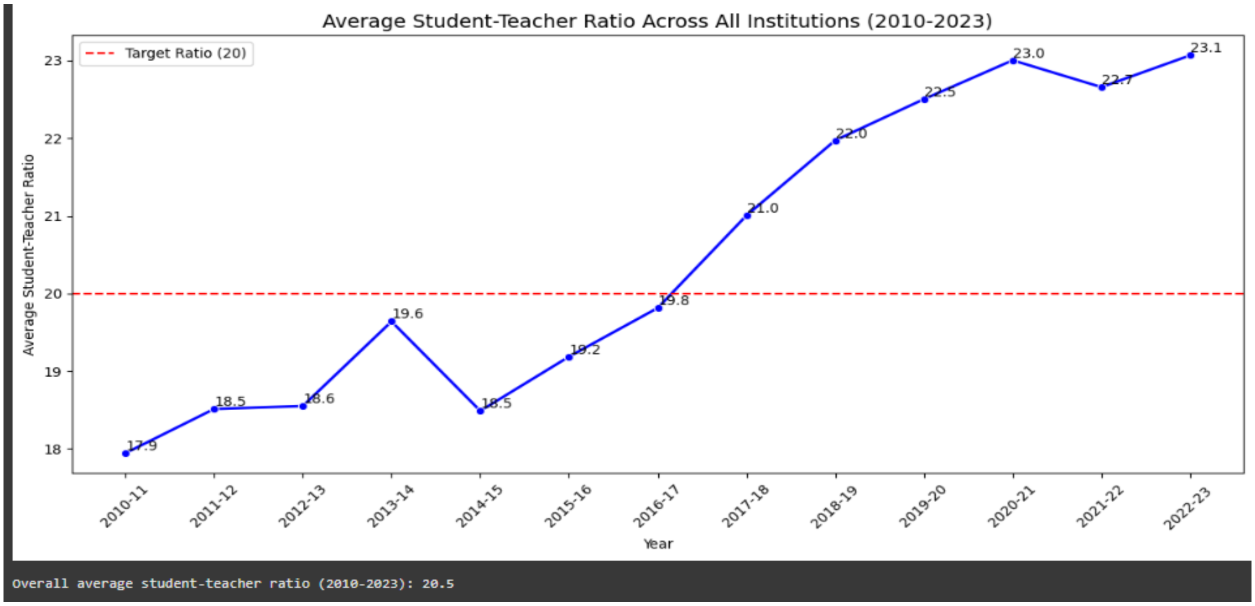
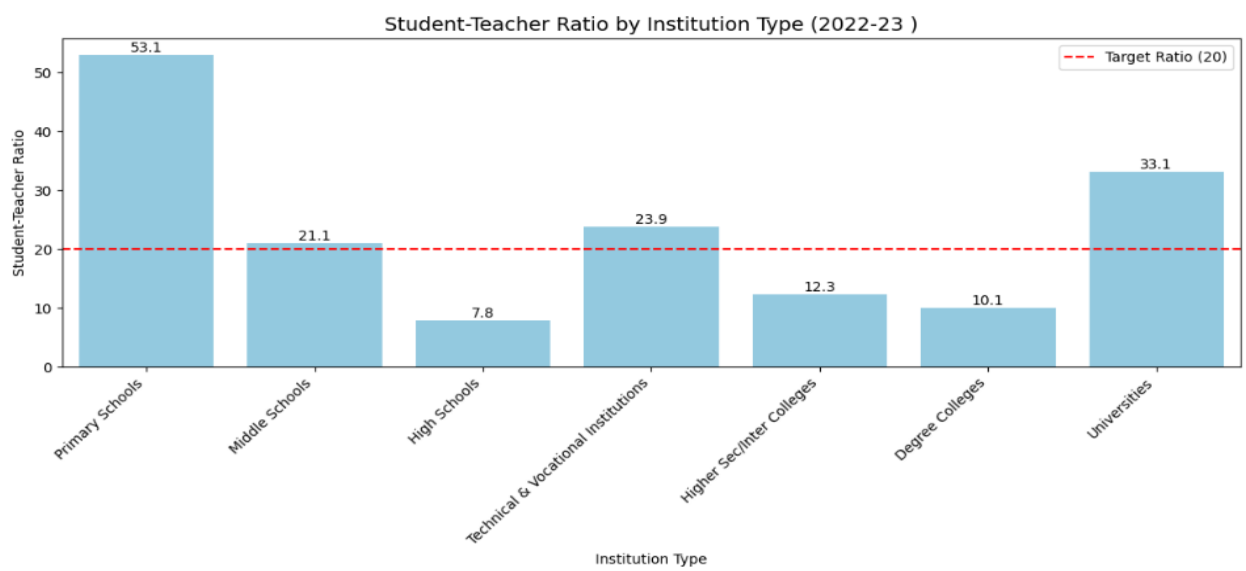
Average Student-Teacher Ratio over the years across all Institutes



Insight: The line chart shows that the average student-teacher ratio across all institutions in Pakistan has consistently increased from 17.9 in 2010-11 to 23.1 in 2022-23, exceeding the target ratio of 20:1 since 2016-17.

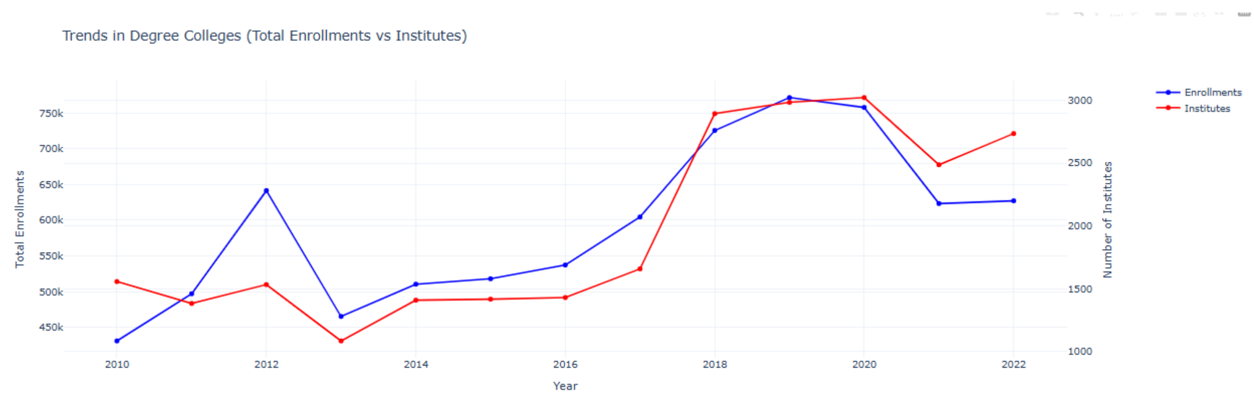
Student-Teacher Ratio by Institutes (Latest)

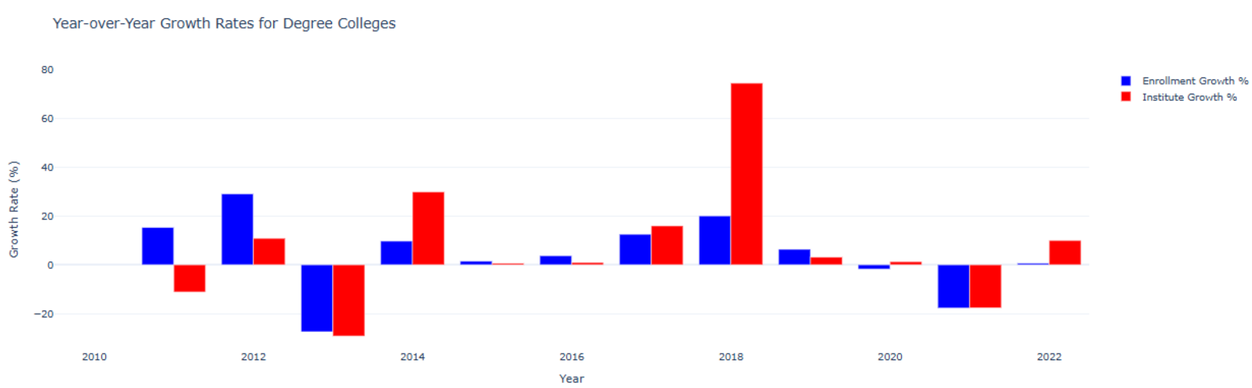


Insight: Primary schools (53.1) and universities (33.1) significantly exceed the target student-teacher ratio of 20:1 in 2022-23.

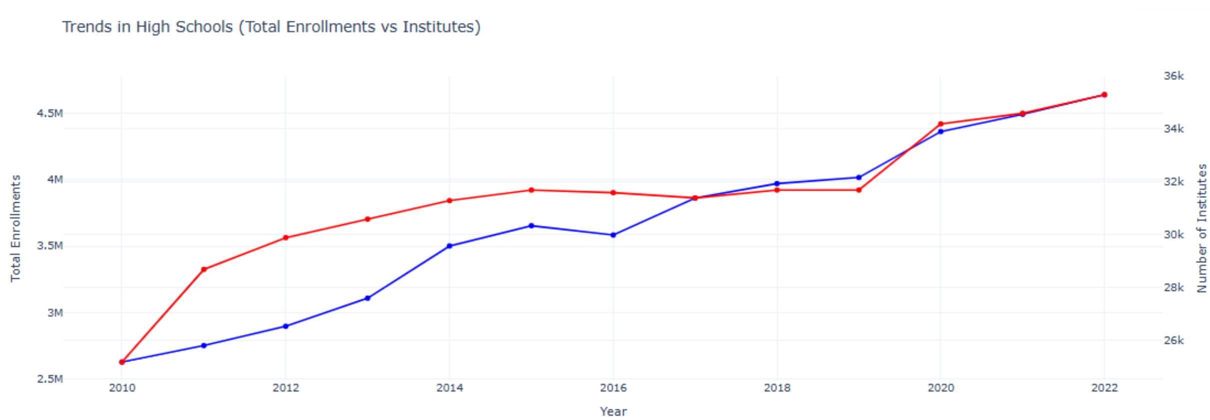
Enrollments vs Institutes

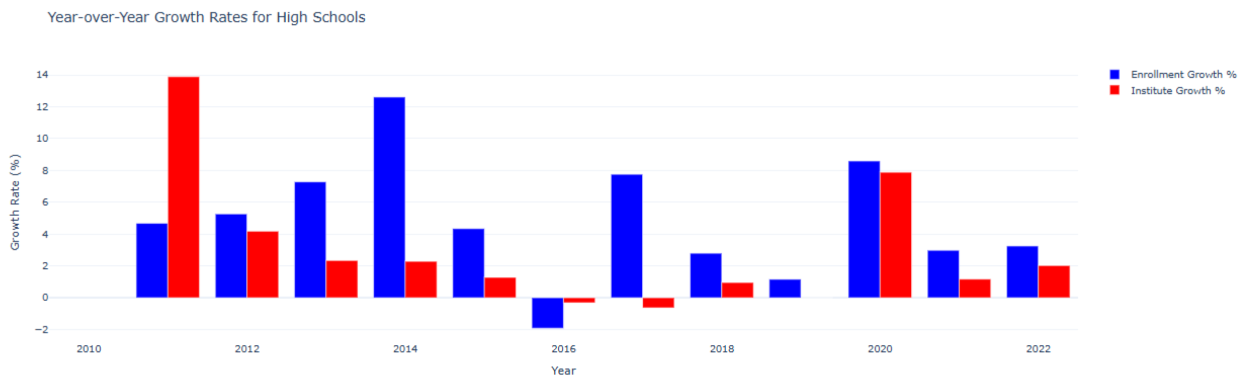
Degree Colleges:



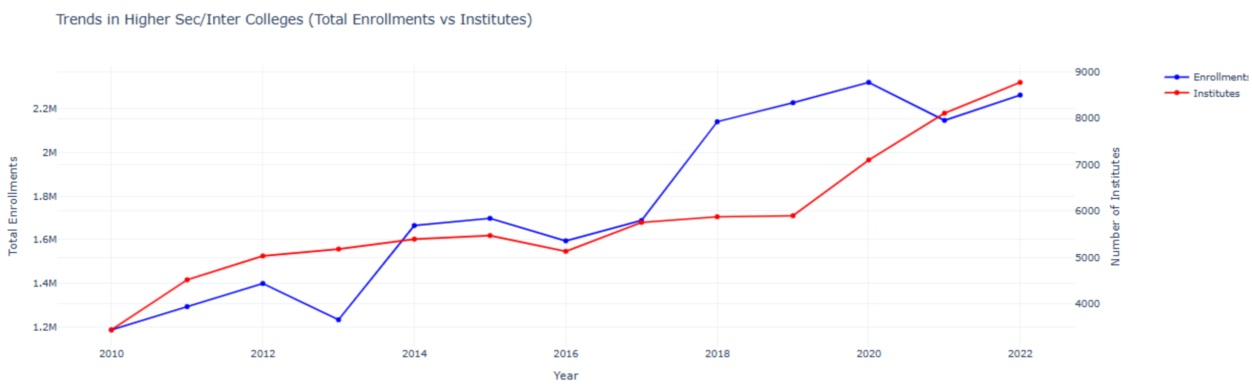


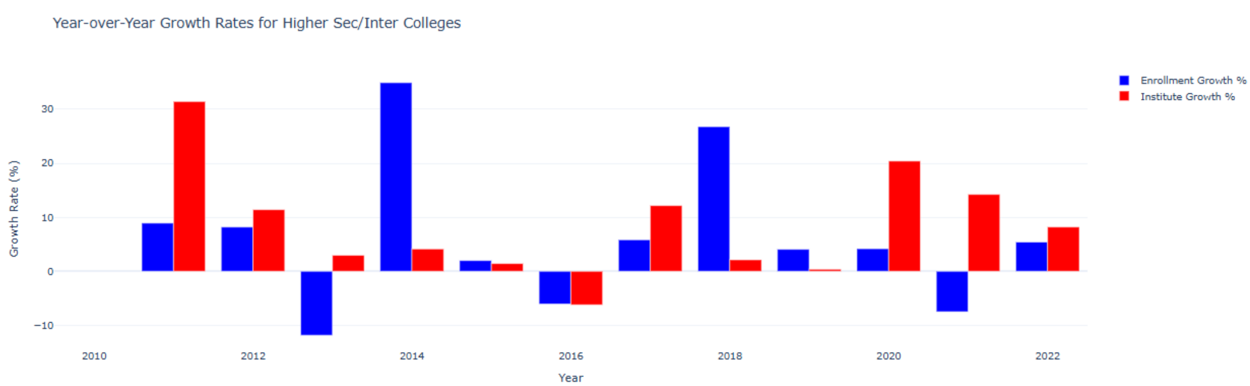
High Schools:



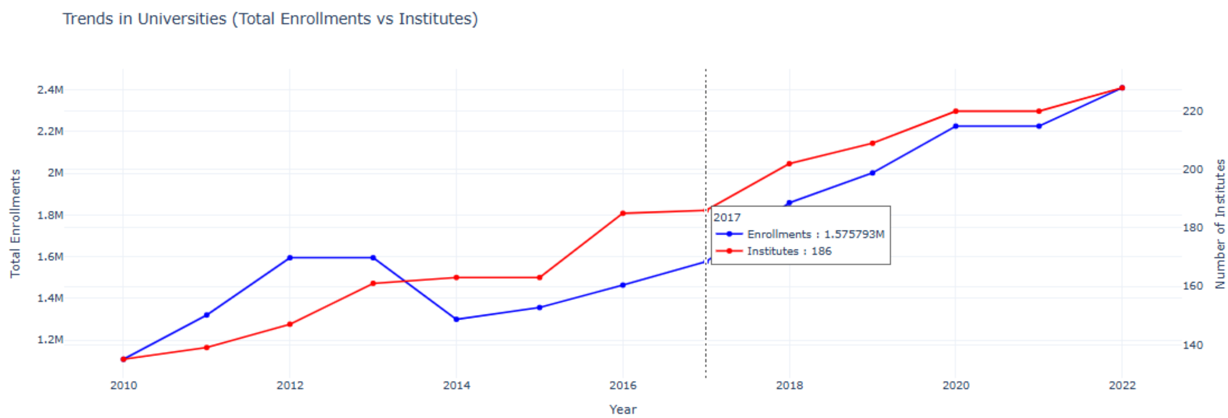


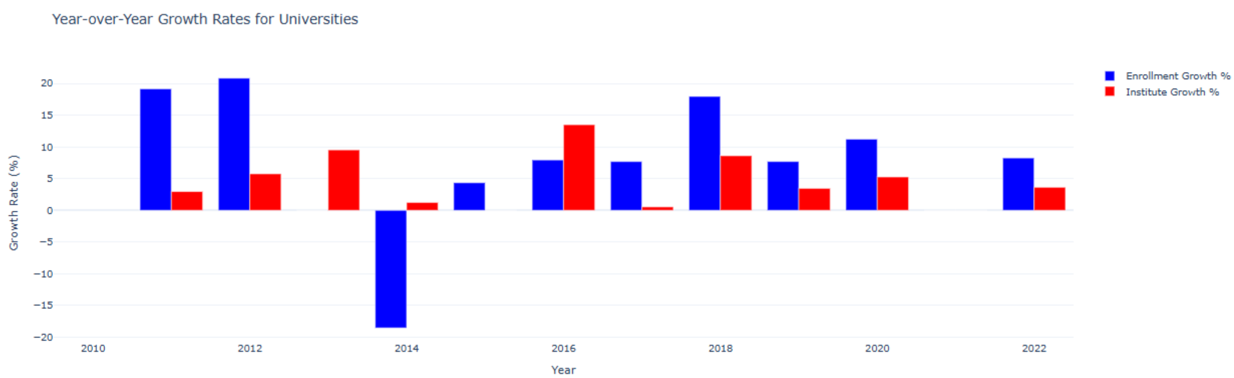
Inter Colleges:



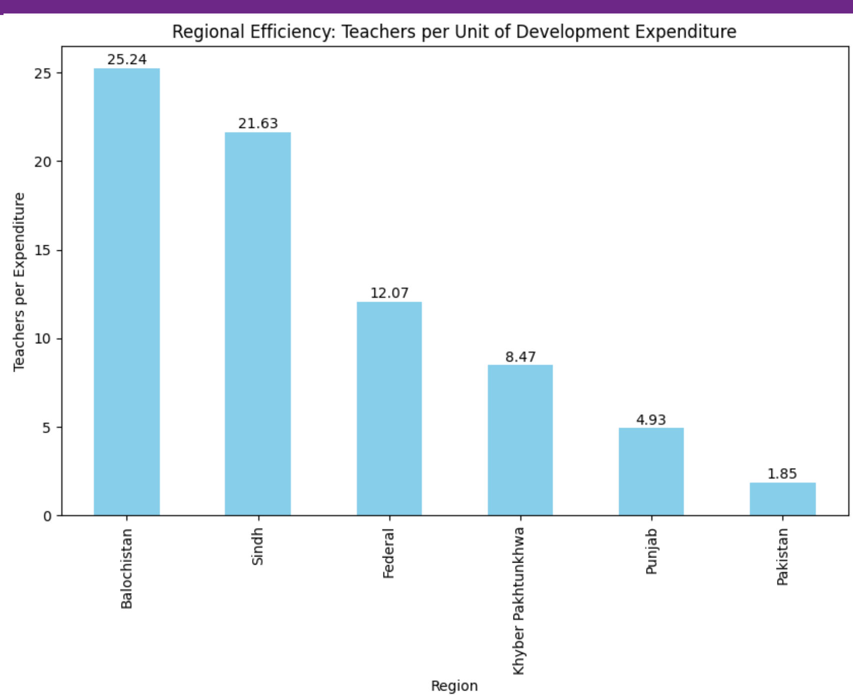


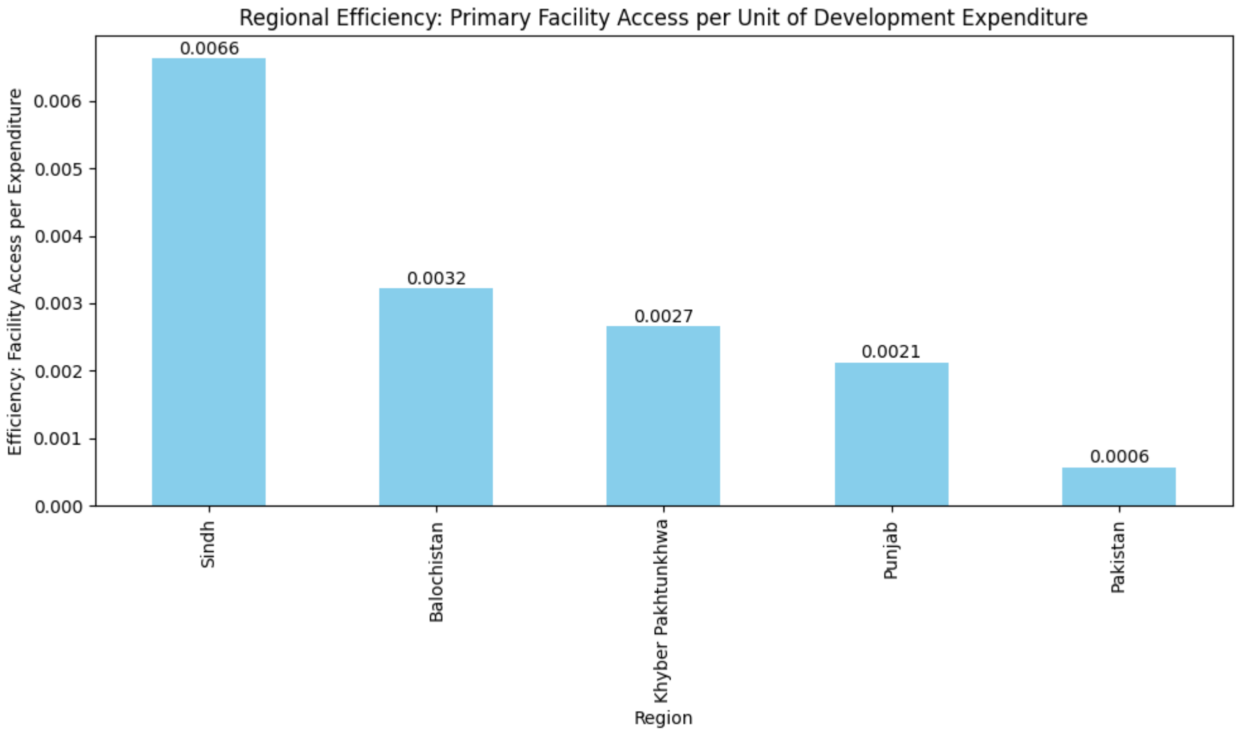
Universities:





Regional Analysis: Impact of Development Expenditure on Facility Access and Teachers Availability





Overall Trends from Enrollments and Institutes:

1. Generally positive growth across all education levels.
2. Higher volatility in higher education compared to school education.
3. Strong correlation between institute growth and enrollment growth.
4. Notable impact of external events (possibly COVID-19) around 2020
5. Recovery and stabilization trends visible in recent years (2020-2022)
6. Middle and high schools showed more consistent growth patterns compared to higher education institutions